

hMPV Model Calibration and Validation using Bounded Nonlinear Least Squares

This document describes the Python implementation used to calibrate and independently validate the seasonally forced hMPV compartmental model for 10 U.S. regions (R1-R10). The workflow includes bounded nonlinear least squares, with a multistart strategy to reduce the risk of retaining a local minimum, fixes the annual seasonal period at 52.18 weeks. It also evaluates practical parameter identifiability by profile likelihood, and performs a 72-week out-of-sample temporal validation without parameter re-estimation.

1. Data Format

Two input CSV files are required for each region:

- Calibration files: R1_python.csv, R2_python.csv, ..., R10_python.csv
- Validation files: R1_validation.csv, R2_validation.csv, ..., R10_validation.csv

Each CSV file contains two columns:

- Column 1: week
- Column 2: case

The case variable is supplied as a proportion between 0 and 1, not as a percentage. Validation observations are reserved exclusively for out-of-sample assessment and are never used for parameter estimation.

2. Model Specification and Fixed Quantities

The state vector is $y(t) = [s_h(t), e_h(t), i_h(t), r_h(t), J(t)]$, where J is an auxiliary cumulative variable used to calculate weekly integrated infectious prevalence. The transmission terms are reparameterized as $\alpha_1 = \kappa_0 \beta_1$ and $\alpha_2 = \kappa_0 \beta_2$. The model has been defined as follows:

$$\begin{aligned} g(t) &= 1 + \kappa_1 \cos\left(\frac{2\pi t}{\omega} + \phi\right) \\ \frac{ds_h}{dt} &= \mu - g(t)(\alpha_1 e_h + \alpha_2 i_h)s_h - \mu s_h + \theta r_h, \\ \frac{de_h}{dt} &= g(t)(\alpha_1 e_h + \alpha_2 i_h)s_h - (\mu + \lambda + v_1)e_h, \\ \frac{di_h}{dt} &= \lambda e_h - (\mu + v_2)i_h, \\ \frac{dr_h}{dt} &= v_1 e_h + v_2 i_h - (\theta + \mu)r_h, \\ \frac{dJ}{dt} &= i_h \end{aligned}$$

The fixed biological parameters are $\mu = 2.5 \times 10^{-4}$, $v_1 = 1.0$, and $v_2 = 0.33$. The seasonal period is fixed for both calibration and validation at $\omega = 52.18$ weeks, corresponding to 365.25/7 weeks per year. The initial state used for calibration is $(S_0, E_0, I_0, R_0, J_0) = (0.97, 0.01, 0.01, 0.01, 0)$.

3. Estimated Parameter Vector

Seven parameters $(\alpha_1, \alpha_2, \kappa_1, \phi, \lambda, \theta, \psi)$ are estimated separately for each region. The first six parameters determine the model dynamics; ψ is an observation-model parameter and does not enter the differential equations. Biologically admissible ranges for all seven parameters have been incorporated.

4. Weekly Observation Model

The model is solved at weekly boundaries with the **DOP853** integrator. The auxiliary state J accumulates $i_h(t)$, so the infectious prevalence integrated over week j is obtained from the difference between

consecutive J values. Because every observation interval has length one week, this integral is numerically equal to the weekly average prevalence.

$$\bar{I}_h^j = J(j) - J(j-1) = \int_{j-1}^j I(t) dt$$

$$\hat{P}_j = \frac{\psi \bar{I}_h^j}{1 - \bar{I}_h^j + \psi \bar{I}_h^j}$$

The quantity $\hat{P}_j$ is therefore the model-predicted hMPV test-positivity proportion corresponding to the observed weekly value.

```
J = solution.y[4, :]
weekly_I = np.diff(J)
weekly_I = np.maximum(weekly_I, 0.0)

psi = p[6]
denominator = 1.0 - weekly_I + psi*weekly_I
predicted_positivity = psi*weekly_I / denominator
```

5. Regional Model Calibration

For each region, the parameters are estimated by minimizing the residual vector between model-predicted and observed positivity. **SciPy least_squares** is used with the **trust-region reflective algorithm** (**method="trf"**) and explicit parameter bounds. The optimization criterion is the sum of squared errors (SSE).

$$r_j(p) = \hat{P}_j(p) - P_j$$

$$SSE(p) = \sum_j [\hat{P}_j(p) - P_j]^2$$

$$\hat{p} = \arg \min_p SSE(p), \text{ subject to } \text{lower} \leq p \leq \text{upper}$$

A multistart search is used to reduce dependence on a single starting point. The first run uses the specified initial guess; subsequent runs use random starting values within biologically admissible ranges. Five independent starts are used for R2-R6 and R8-R10, whereas R1 and R7 use 20 starts. The solution with the smallest SSE is retained.

```
result = least_squares(
    residual_function,
    x0=x0,
    args=(observed_data,),
    bounds=(lower_bounds, upper_bounds),
    method="trf",
    x_scale="jac",
    ftol=1e-10,
    xtol=1e-10,
    gtol=1e-10,
    max_nfev=5000
)

sse = np.sum(result.fun**2)
# retain the run with the smallest SSE
```

6. Calibration Goodness-of-Fit and Reproduction-Number Summaries

After calibration, the fitted trajectory system is re-evaluated and the following statistics are calculated: SSE, MSE, RMSE, MAE, NRMSE, R^2 , AIC, and BIC. NRMSE is defined by dividing RMSE by the observed data range. The code also reports baseline, minimum-season, and maximum-season reproduction-number summaries.

$$R_{0,base} = \frac{\alpha_1}{\mu + \lambda + \nu_1} + \frac{\alpha_2 \lambda}{[(\mu + \lambda + \nu_1)(\mu + \nu_2)]}$$

$$R_{0,min} = (1 - \kappa_1)R_{0,base}, \quad R_{0,max} = (1 + \kappa_1)R_{0,base}$$

Calibration residuals are stored as observed minus fitted positivity, and the script exports region-specific fitted-data files together with calibration, residual, state-variable, and seasonal-forcing figures.

7. Profile-Likelihood Analysis

For a selected parameter p_i , the parameter is fixed at a sequence of profile values and the remaining six parameters are re-optimized. Thirty-one profile points are used for each parameter (15 on each side of the optimum plus the optimum). Positive parameters spanning broad ranges are profiled on a logarithmic grid, whereas κ_1 and ϕ use linear spacing.

$$SSE_{i,\text{prof}}(c) = \min_{p_{-i}} SSE(p_i = c, p_{-i})$$

$$LR_i(c) = N \log[SSE_{i,\text{prof}}(c) / SSE_{\min}]$$

A profile value belongs to the asymptotic 95% likelihood-ratio confidence set when $LR_i(c) \leq \chi^2(1, 0.95) = 3.84146$. If the profile does not cross the threshold before reaching an imposed parameter bound, the corresponding confidence interval endpoint is reported as bound-limited rather than as a fully finite likelihood interval.

8. Independent 72-Week Temporal Validation

Validation is performed strictly after calibration and without refitting. The fitted terminal biological state $(s_h^{\text{end}}, e_h^{\text{end}}, i_h^{\text{end}}, r_h^{\text{end}})$ is used to initialize the validation simulation. The auxiliary cumulative variable J is reset to zero because it is used only to integrate $i_h(t)$ within the future weekly intervals.

```
validation_y0 = np.array([
    terminal_state[0], # S_end
    terminal_state[1], # E_end
    terminal_state[2], # I_end
    terminal_state[3], # R_end
    0.0                # reset J
])

t_start = float(model_time_end)
t_end = t_start + float(n_validation)
# same fitted parameters p; no re-estimation
```

The validation predictions are compared against the 72 reserved observations using SSE, MSE, RMSE, MAE, NRMSE, and R^2 . Residuals are defined as observed minus predicted positivity. A separate validation-prediction CSV is saved for every region.

9. Regional Processing Workflow

For each of the 10 regions, the script executes the following sequence:

1. Load and verify the calibration and 72-week validation datasets.
2. Estimate the seven regional parameters by bounded multistart nonlinear least squares.
3. Re-evaluate the best-fit trajectory and compute calibration diagnostics.
4. Evaluate the terminal state at the end of calibration.
5. Generate the independent 72-week prediction without parameter refitting.
6. Compute validation diagnostics and export observed/predicted values.
7. Construct seven profile-likelihood curves and 95% profile confidence intervals.
8. Save regional figures, profile data, parameter tables, and forecast-initialization information.

10. Main Output Files

Important combined outputs include:

- tables/Table_parameter_estimates_95CI_long.csv
- tables/Table_parameter_estimates_95CI_wide.csv
- tables/Table_profile_CI_status_all_regions.csv
- tables/Table_goodness_of_fit_all_regions.csv
- tables/Table_multistart_fits_all_regions.csv
- tables/Table_forecast_initial_conditions_all_regions.csv

- tables/Table_validation_metrics_all_regions.csv
- combined_figures/All_Regions_Calibration_Validation.png/.pdf/.svg
- combined_figures/All_Regions_Residuals.png/.pdf/.svg
- combined_figures/All_Regions_Seasonal_Forcing.png/.pdf/.svg
- supplementary_profiles/R1_profile_likelihoods ... R10_profile_likelihoods (.png/.pdf/.svg)

In addition, each regional directory contains the fitted-data CSV, validation-prediction CSV, parameter estimates with 95% profile intervals, profile-data CSV files, and publication-quality calibration, residual, state-variable, seasonal-forcing, and profile-likelihood figures.

The program checks for missing input files before fitting begins, processes R1-R10 sequentially, writes checkpoint tables after each completed region, and then generates the combined all-region tables and figures.